Quantification: 699_HYP_HP_z11c1-2
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	30
interior lacuna count	25
frame-edge lacuna count	5

Interior lacunae only: mean, median and sample SD (n = 25)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	1946.6	971	2669.63	px^2
major_axis_length_px	lacuna shape	72.9851	54.0186	48.0087	px
minor_axis_length_px	lacuna shape	35.3197	30.4091	19.8543	px
aspect_ratio	lacuna shape	2.1706	2.0007	0.8922	unitless
eccentricity	lacuna shape	0.8087	0.8661	0.1461	unitless
solidity	lacuna shape	0.7102	0.7052	0.0991	unitless
roots_count	headline	8.36	8	5.1062	count
ring_length_r30_px	headline	295.32	275	188.705	px
ring_length_r60_px	reported	515.64	488	356.486	px
owned_length_px	ownership-dependent	594.101	412.706	659.155	px
edge_count	ownership-dependent	32.16	24	36.5817	count
mean_edge_length_px	ownership-dependent	18.258	17.8995	7.7224	px
perimeter_px	normalised	272.165	182.894	206.596	px
ring_area_r30_px2	normalised	7568.08	6958	4485.74	px^2
ring_area_r60_px2	normalised	16029.5	14086	9239.28	px^2
in_frame_fraction_r30	normalised	0.9959	1	0.017	unitless
in_frame_fraction_r60	normalised	0.9863	1	0.0474	unitless
ring_density_r30	normalised	0.0393982	0.0427179	0.0159429	px^-1
ring_density_r60	normalised	0.0318751	0.0340635	0.0138127	px^-1
roots_per_100px_perimeter	normalised	3.5012	3.4797	1.9222	per 100 px
ring_attached_length_r30_px	network v2	253.2	245	188.514	px
ring_attached_length_r60_px	network v2	384.16	380	331.981	px
ring_length_w_r30_px	network v2	332.142	308.137	211.514	px
ring_length_w_r60_px	network v2	577.375	547.688	397.309	px
sholl_crossings_r10	network v2	6.8	6	4.8477	count
sholl_crossings_r20	network v2	7.28	6	4.3639	count
sholl_crossings_r30	network v2	7.08	7	4.4527	count
loop_count	loops, report only	1.64	0	3.2645	count
loop_edge_share	loops, report only	0.1042	0	0.1342	share
ring_width_mean_r30_px	width	7.5405	7.7118	1.319	px

Per field
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measure	value	units
analysed_area_px2	961876	px^2
total_skeleton_length_px	21874	px
canalicular_length_density_per_px	0.022741	px^-1
junction_count	481	count
junction_density_per_px2	0.00050006	px^-2
lacuna_count	30	count

lacunae_per_px2	3.119e-05	px^-2
skeleton_component_count	333	count
mean_component_length_px	65.6877	px
median_component_length_px	15	px
field_length_density_w_per_px	0.0253829	px^-1
field_density_without_flagged_per_px	0.022051	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	603	px
unattached_component_count	31	count
canalicular_length_density_attached_per_px	0.022741	px^-1
field_length_density_with_unattached_per_px	0.0233679	px^-1
loop_count	54	count
loop_edge_share	0.1677	share
edge_count	1282	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.5609	px
width_median_px	8	px
width_p10_px	4	px
width_p90_px	10.7703	px
width_pixels_used	17109	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	483	34.5143	18.5791	1.8577	0.8428	0.9028	-1.3729	9.6004	721.977	True
2	889	60.8664	21.5104	2.8296	0.9355	0.7566	-1.1858	41.9595	618.117	False
3	971	48.2689	39.9508	1.2082	0.5612	0.6637	-1.1736	126.652	613.53	False
4	1293	67.1123	29.3143	2.2894	0.8996	0.7085	-0.9693	147.124	738.418	False
5	933	52.1712	30.6068	1.7046	0.8098	0.7272	-0.9318	146.222	325.69	False
6	439	27.0007	22.5757	1.196	0.5486	0.8591	-1.0184	203.804	191.597	False
7	437	44.3733	15.1685	2.9254	0.9398	0.7083	1.2486	210.419	385.556	False
8	8385	126.932	101.718	1.2479	0.5982	0.7052	-1.4611	277.754	204.132	False
9	1905	74.7758	58.1104	1.2868	0.6293	0.5641	1.396	301.221	621.288	False
10	657	45.9581	30.4091	1.5113	0.7498	0.5516	0.7816	290.753	563.455	False
11	9083	140.536	109.35	1.2852	0.6281	0.6544	-0.1917	390.607	39.1286	True
12	11096	170.603	105.697	1.6141	0.785	0.7494	-0.1521	533.955	42.6183	True
13	1403	54.0186	34.896	1.548	0.7633	0.888	-0.1596	368.566	862.24	False
14	3093	92.0714	54.1062	1.7017	0.8091	0.6916	1.2856	410.259	393.304	False
15	1944	82.0911	39.6724	2.0692	0.8755	0.6506	1.2002	562.526	659.101	False
16	455	30.8298	20.9014	1.475	0.7351	0.8228	1.2448	565.978	720.266	False
17	1195	44.9336	38.8022	1.158	0.5043	0.7607	-1.4455	562.059	755.791	False
18	2018	112.423	30.231	3.7188	0.9632	0.6731	1.4432	571.53	456.488	False
19	634	48.158	30.1352	1.5981	0.78	0.5992	-1.5103	631.199	551.279	False
20	1329	62.4213	28.6247	2.1807	0.8887	0.8819	1.2484	643.479	606.14	False
21	590	46.6483	23.3159	2.0007	0.8661	0.6095	-1.4011	672.297	622.215	False
22	812	43.7603	30.951	1.4139	0.7069	0.7167	-1.4386	655.233	193.442	False
23	451	43.3593	20.67	2.0977	0.8791	0.6499	-0.8882	692.12	316.188	False
24	587	59.9418	19.0149	3.1524	0.9484	0.5959	-1.5348	698.601	352.387	False

25	12055	251.462	78.5981	3.1993	0.9499	0.6372	-0.5972	767.904	745.585	False
26	451	45.8335	13.8387	3.312	0.9533	0.8623	0.6712	773.745	871.692	False
27	2798	118.207	35.7656	3.305	0.9531	0.8127	-1.3169	814.447	954.982	False
28	2396	80.3638	41.3693	1.9426	0.8573	0.8812	-1.0185	852.972	994.821	True
29	14977	374.03	83.4021	4.4847	0.9748	0.5035	-1.4304	910.923	184.857	True
30	2941	141.011	34.1041	4.1347	0.9703	0.6575	1.2599	895.57	423.196	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	1	98	161	41.1421	1	41.1421	5.5126
2	4	203	249	171.569	11	15.5971	7.3657
3	10	358	604	725.027	44	16.4779	8.6418
4	12	386	697	821.826	35	23.4807	8.1517
5	8	298	437	412.706	26	15.8733	9.642
6	6	142	243	90.2843	9	10.0316	8.3226
7	7	222	488	457.505	23	19.8915	9.0224
8	18	742	1280	1556.99	92	16.9238	7.7584
9	15	385	781	811.056	56	14.4831	7.1094
10	10	275	551	605.676	35	17.305	8.2415
11	17	546	744	648.772	50	12.9754	8.1185
12	10	245	370	335.978	18	18.6654	8.7693
13	2	209	470	315.622	15	21.0415	6.8106
14	18	813	1542	3029.25	169	17.9246	7.4887
15	11	397	533	652.872	35	18.6535	7.7118
16	0	10	10	0	0	0	10.3389
17	2	121	225	17.8995	1	17.8995	4.7611
18	13	424	731	726.399	35	20.7543	7.4846
19	11	234	311	389.291	24	16.2205	6.3077
20	2	174	186	33.3137	3	11.1046	8.1266
21	3	160	225	140.154	3	46.7181	5.9762
22	10	360	596	789.713	37	21.3436	8.5944
23	7	206	372	405.363	20	20.2681	7.4959
24	6	276	556	727.701	36	20.2139	8.1101
25	12	409	639	312.078	13	24.006	5.9067
26	5	116	139	99.2548	6	16.5425	5.2586
27	3	31	102	25.8995	2	12.9497	6.0131
28	0	0	0	0	0	0	
29	20	603	1219	1321.97	82	16.1216	8.5339
30	14	432	924	1535.08	74	20.7444	7.8714

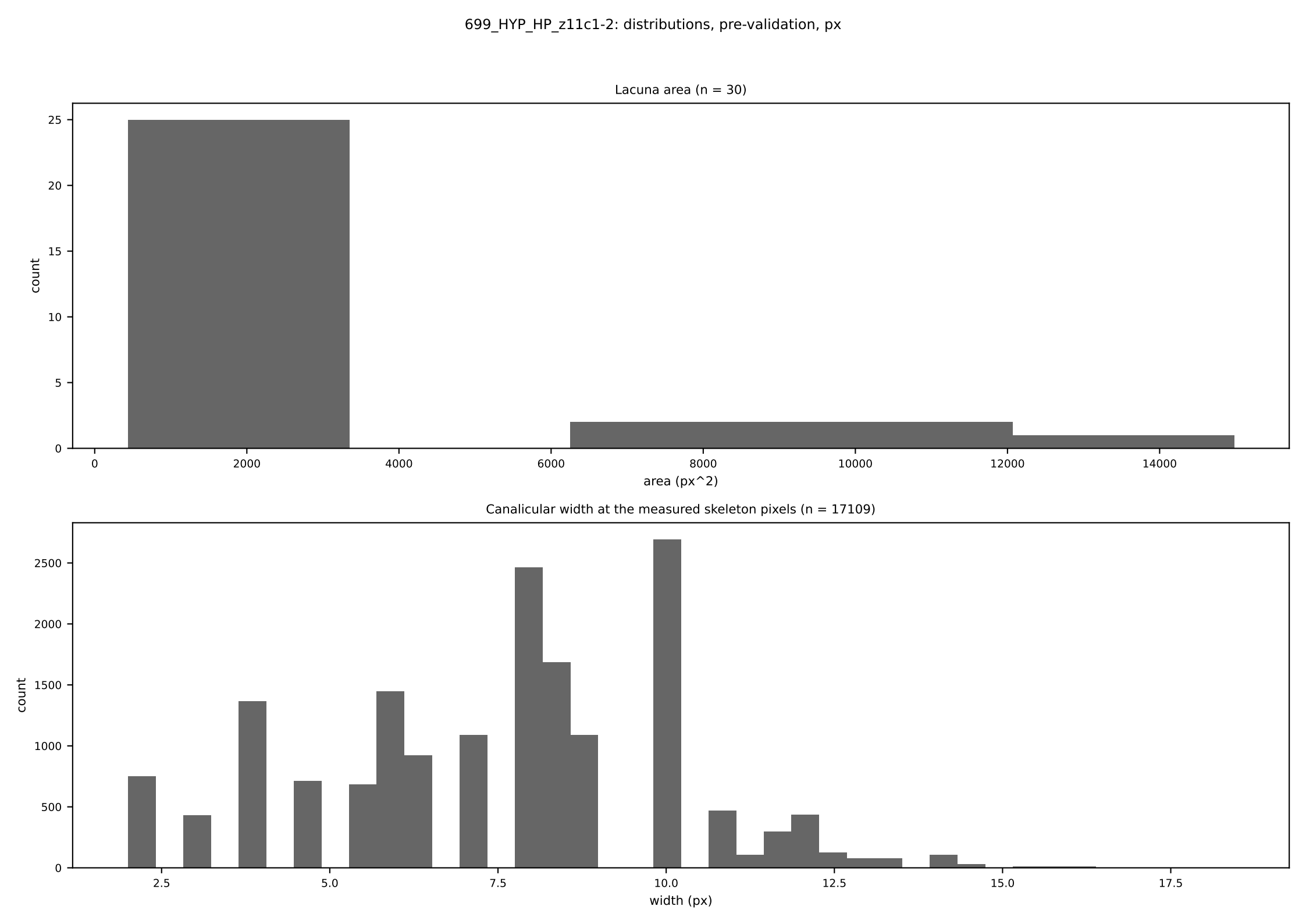
Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	95.2548	3348	7313	0.6198	0.5747	0.0292712	0.0220156	1.0498
2	172.995	6815	11086	0.9811	0.8306	0.0297872	0.0224608	2.3122
3	191.622	7249	15098	1	1	0.0493861	0.0400053	5.2186
4	234.25	8068	19697	1	1	0.0478433	0.0353861	5.1227
5	172.995	6976	15405	1	1	0.0427179	0.0283674	4.6244
6	91.4975	3556	8201	1	1	0.0399325	0.0296305	6.5576
7	120.569	5863	14086	1	1	0.0378646	0.0346443	5.8058
8	894.833	15496	28319	1	1	0.0478833	0.0451993	2.0115
9	394.333	8052	18251	1	1	0.0478142	0.0427922	3.8039
10	186.066	5049	11971	1	1	0.0544662	0.0460279	5.3744
11	756.861	10512	16869	0.6862	0.6501	0.0519406	0.0441046	2.2461
12	695.896	10926	20640	0.7085	0.658	0.0224236	0.0179264	1.437
13	174.066	7351	20270	1	1	0.0284315	0.023187	1.149

14	467.446	11796	28229	1	1	0.0689217	0.0546247	3.8507
15	318.333	8398	15415	1	1	0.0472732	0.0345767	3.4555
16	104.468	1694	2264	1	1	0.00590319	0.00441696	0
17	232.285	5734	14018	1	1	0.0211022	0.0160508	0.861
18	409.931	10384	22846	1	1	0.0408321	0.0319969	3.1713
19	174.853	5189	9130	1	1	0.0450954	0.0340635	6.291
20	172.267	2782	2935	1	1	0.0625449	0.0633731	1.161
21	155.196	4616	9759	1	1	0.034662	0.0230556	1.933
22	182.894	6958	18413	1	1	0.051739	0.0323684	5.4676
23	129.539	4397	10895	1	1	0.0468501	0.0341441	5.4038
24	172.439	5201	12627	1	1	0.0530667	0.0440326	3.4795
25	834.833	23174	45500	1	1	0.0176491	0.014044	1.4374
26	109.296	4791	7177	1	1	0.0242121	0.0193674	4.5747
27	304.777	7140	12831	0.9164	0.8272	0.00434174	0.0079495	0.9843
28	235.48	4021	7954	0.6927	0.6449	0	0	0
29	1353.43	28693	52942	0.9016	0.8594	0.0210156	0.0230252	1.4777
30	402.333	12473	26315	1	1	0.0346348	0.035113	3.4797

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	85	99	105.184	173.64	4	2	2
2	99	103	225.924	274.38	5	6	8
3	329	496	413.99	689.713	8	8	10
4	374	621	434.049	768.115	9	12	12
5	268	388	345.492	503.718	6	9	6
6	96	96	156.326	274.48	3	3	3
7	218	380	252.551	547.688	6	6	6
8	718	1062	826.541	1424.12	21	13	12
9	362	604	429.948	864.269	9	8	11
10	248	461	308.137	608.475	5	8	6
11	513	595	620.215	850.352	12	14	8
12	233	277	278.108	422.061	7	7	2
13	169	284	233.752	529.76	4	4	7
14	799	1489	916.37	1721.76	18	18	17
15	322	402	445.12	600.345	10	11	9
16	10	10	12.0711	12.0711	1	0	0
17	47	47	132.811	244.581	3	4	4
18	400	491	468.605	820.754	11	13	7
19	206	223	262.894	347.007	5	5	4
20	166	166	199.681	215.167	4	3	2
21	118	142	175.953	242.823	4	4	4
22	321	490	406.563	668.286	8	11	9
23	194	296	238.894	424.848	4	6	4
24	254	489	309.794	623.316	7	5	8
25	245	294	461.274	727.767	8	11	8
26	70	71	135.297	159.125	1	4	2
27	26	26	31.8995	112.669	0	0	1
28	0	0	0	0	0	0	0
29	438	1131	656.103	1351.76	15	20	15
30	271	473	479.605	1029.15	10	10	17


699_HYP_HP_z11c1-2: distributions, pre-validation, px
Lacuna area (n = 30)
count
area (px^2)
0
2000
4000
6000
8000
10000
12000
14000
0
5
10
15
20
25
Canalicular width at the measured skeleton pixels (n = 17109)
count
width (px)
2.5
5.0
7.5
10.0
12.5
15.0
17.5
0
500
1000
1500
2000
2500